

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2118

Warminster 138 kV
June 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2118. The Interconnection Customer proposes a 109.0 MW battery energy storage facility interconnecting at the Warminster 138 kV substation in the Pennsylvania area. The request was received on June 18, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 468 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,185,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Albany - Thetford 138.0 kV	Pennsuco - Cherokee Falls 138.0 kV	112.4	212	20.4
T Bone 7011 - Portside Energy 138.0 kV	Warrington Substation - Sunbeam 138.0 kV	110.1	268	50.9
Pennsuco - Cherokee Falls 138.0 kV	Elroy - Rothadew 138.0 kV	109.4	297	10.5
Pennsuco - Whitpain 138.0 kV	Elroy Substation - Rothadew 138.0 kV	107.2	279	10.5
Sunbeam - Warrington 138.0 kV	Portside Energy Substation - T Bone 701 138.0 kV	104.0	273	51.0
Sunbeam - Mcgen 138.0 kV	T Bone 7011 Substation - Portside Energy Substation 138.0 kV	103.3	244	51.0
Medway - Petrie 138.0 kV	Ganade - Jewell Branch 138.0 kV	103.1	233	28.8
Wallenaupack Substation 500-345 kV Auto	West Gallatin - Walnut 345.0 kV	100.2	2,125	7.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Albany - Thetford 138.0 kV	Rebuild line with higher-capacity conductor	\$1,800,000	100% (\$1,800,000)
Portside Energy - T Bone 7011 138.0 kV	Rebuild line with higher-capacity conductor	\$949,000	100% (\$949,000)
Cherokee Falls - Pennsuco 138.0 kV	Reconductor line	\$347,000	100% (\$347,000)
Pennsuco - Whitpain 138.0 kV	Reconductor line	\$2,739,000	100% (\$2,739,000)
Warrington - Sunbeam 138.0 kV	Reconductor line	\$694,000	100% (\$694,000)
Sunbeam - Mcgen 138.0 kV	Reconductor line	\$292,000	100% (\$292,000)
Medway - Petrie 138.0 kV	Reconductor line	\$2,100,000	100% (\$2,100,000)
Wallenaupack Substation 500-345 kV Auto	Replace transformer with higher-rated unit	\$29,540,000	1% (\$264,000)
	Total allocated to project		\$9,185,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.